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1. Let $X \sim N(\mu, \Sigma)$ be n dimensional, what is the distribution of $(X-\mu)^t A(X-\mu)$ for a symmetric matrix A of rank k?

1 / 1 point

Correct

2. Consider a linear model $Y \sim N(X\beta, \sigma^2 I)$. Let k be a $p \times 1$ vector of contrasts and let s be the residual standard deviation. What can be said about:

0.75 / 1 point

$$\left(\frac{k^t\hat{\beta}-m}{\sqrt{s/(k^t(X'X)^{-1}k)}}\right)^2$$

(Check all that apply.)

Incorrect

3. Let e_i be residual i from a linear model under the assumption that $Y \sim N(X\beta, \sigma^2 I)$. What can be said about

1 / 1 point

$$\frac{e_i}{s\sqrt{1-h_{ii}}}$$

where $var(e_i) = (1-h_{ii})\sigma^2$ and s is the residual standard deviation.

Correct

4. In a prediction interval where we want to predict at $p \times 1$ vector x_{new}, if we collect data so that $x_{new}^t(X^tX)^{-1}x_{new}$ goes to zero as $n \to \infty$ then our prediction interval will converge to a point at the true value?

1 / 1 point

Correct

5. In a confidence interval for $x_{new}^t\beta$ if we collect data so that $x_{new}^t(X^tX)^{-1}x_{new}$ goes to zero as $n \to \infty$ then our confidence interval will converge to a point at the true value?

1 / 1 point

Correct

6. Let K be a full row rank $2 \times p$ matrix consider the model $Y \sim N(X\beta, \sigma^2 I)$, null hypothesis $H_0 : K\beta = m$, and test statistic

0.6 / 1 point

$$0.5(K\hat{\beta}-m)^t(K(X^tX)^{-1}K^tS^2)^{-1}(K\hat{\beta}-m)$$

where s is the residual variance estimate. Which of the following are true? (Check all that apply.)

Incorrect

7. Let $e_{i,-i} = Y_i - \hat{Y}_{i,-i}$ be residual i from a linear model under the assumption that $Y \sim N(X\beta, \sigma^2 I)$ where $\hat{Y}_{i,-i}$ refers to the predicted value for subject i where the model was fit with the i^{th} data point removed. Let S_{-i} be the estimated residual standard deviation obtained from the model fit with the i^{th} data point removed. Consider the statistic:

1 / 1 point

$$\frac{e_{i,-i}}{S_{-i}\sqrt{1-h_{ii}}}$$

where $var(e_{i,-i}) = (1-h_{ii})\sigma^2$. What can be said about this statistic (check all that apply)?

Correct